

UNIT 5 · TOPIC 5.11

Overview of Emerging Technologies

Case studies from the edge of aviation tech.



B.B.A. Aviation Management · Semester III · Unit 5: Technology & Trends
Air Ticketing & Fare Construction (2ACBAM32) · Arun Kumar

TOPIC 5.11

Mission Video Briefing



Air Ticketing & Fare Construction · Unit 5: Technology & Trends · Topic 5.11



02

TOPIC 5.11

Mission Question



THINK ABOUT IT

Which will change airports most: 5G, drones, biometrics or blockchain?

**THINK**

30 seconds alone

**PAIR**

1 minute with a neighbour

**SHARE**

2-3 answers with the class



TOPIC 5.11

Mission Briefing

1

5.11.1

The role of 5G and IoT in transforming travel

2

5.11.2

Drones for baggage handling and ticketing

3

5.11.3

Smart airports and biometric verification

4

5.11.4

Blockchain and smart contracts in air ticketing

**MISSION OBJECTIVES**

- Explain 5G/IoT applications
- Assess drone use realistically
- Describe smart airport integration
- Analyse smart-contract ticketing

Pedagogy: Case Study-Based Learning

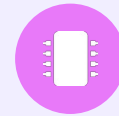
TOPIC 5.11.1

5G & IoT in Travel



5G Speed

Fast, low-latency data for apps and airport systems.



IoT Sensors

Track bags, carts, aircraft and queues.



Smart Bags

RFID/Bluetooth tags report location.



Live Dashboards

Queue times shown to passengers.



TOPIC 5.11.1

Case: IoT Baggage Tracking

**RFID tags**

Bags read automatically at key points (IATA Res 753).

**Passenger app**

Shows where your bag is in real time.

**Fewer mishandled bags**

Problems spotted before the bag is lost.

**CASE QUESTION**

An airline's mishandled bag rate is high at one hub. How could RFID + IoT data help find the root cause?



TOPIC 5.11.2

Drones: Reality Check



Real Uses Today

- Aircraft inspections
- Airport perimeter security
- Cargo delivery trials
- Runway/infrastructure checks

VS



Experimental / Future

- Baggage movement
- Delivering items to passengers
- Indoor inventory checks
- Strict safety rules limit use near aircraft



TOPIC 5.11.2

Case Study Method

1

Situation

2

Problem

3

Options

4

Decision & lessons

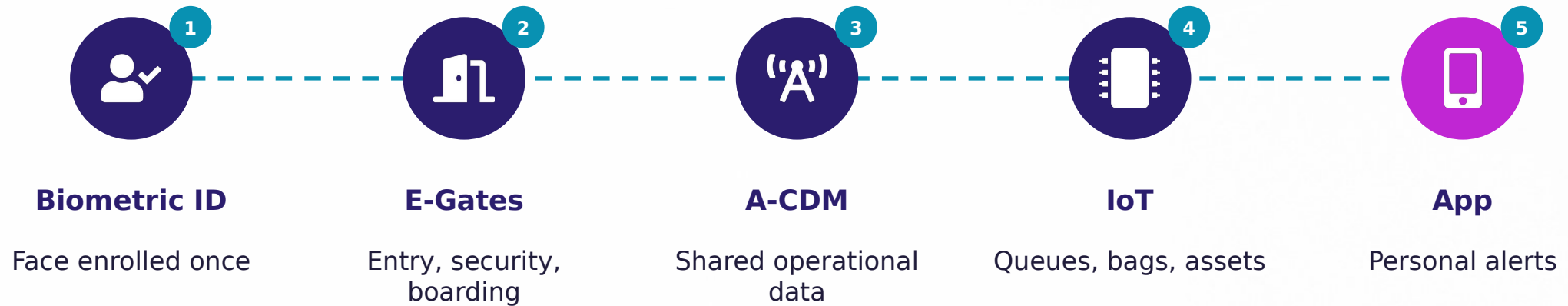


Use this 4-step structure for every case in the activity.



TOPIC 5.11.3

Smart Airport Integration



TOPIC 5.11.3

Case: DigiYatra



How

Upload Aadhaar-linked ID + selfie in app; boarding pass linked.



Result

Faster entry and boarding without paper.



Debate

Data storage and privacy questions discussed publicly.

TOPIC 5.11.4

Case: Smart-Contract Flight Delay Cover

Insurance that pays itself (concept based on real pilots).

- Passenger buys delay cover with ticket.
- Contract reads official flight data.
- If delay exceeds threshold, payout triggers.
- No claim form needed.

SMART CONTRACT (concept)

```
IF flight.delay >= 120 min  
THEN pay(passenger, INR 2000)  
  
DATA SOURCE: flight status feed  
STATUS: AI887 delayed 145 min  
  
-> PAYOUT SENT
```

TOPIC 5.11.4

Case Comparison

Technology	Maturity	Main benefit	Main barrier
5G / IoT	Growing	Real-time tracking	Infrastructure cost
Drones	Early	Inspections, security	Safety regulation
Biometrics	Deployed	Speed, no paper	Privacy concerns
Smart contracts	Pilots	Automatic payouts	Adoption, data trust

TOPIC 5.11

Innovation Lab: Tech Case Tribunal



PEDAGOGY

Case Study-Based Learning

TIME · 15 min

FORMAT · 4 case groups

- 1 Each group analyses one case: IoT bags, drones, DigiYatra, smart contracts.
- 2 Apply the 4-step case method.
- 3 Decide: invest now, pilot, or wait — with reasons.
- 4 Class “tribunal” questions each group’s decision.



TOPIC 5.11

Click to reveal each answer

System Check Quiz

Q1**IATA Resolution 753 relates to...**

- A) Baggage tracking
- B) Fare rules
- C) Taxes
- D) Refunds

ANSWER

A**Q2****Drones at airports are mainly used today for...**

- A) Carrying passengers
- B) Inspections and security
- C) Selling tickets
- D) Catering

ANSWER

B**Q3****A smart contract pays out...**

- A) After manual claims only
- B) Automatically when conditions are met
- C) Never
- D) Monthly

ANSWER

B**Q4****The main concern with biometrics is...**

- A) Speed
- B) Privacy
- C) Colour
- D) Noise

ANSWER

B

MISSION COMPLETE

Key Takeaways

- ✓ 5G/IoT enable real-time tracking and smarter airports
- ✓ Drones: real for inspections, experimental for bags
- ✓ Smart airports integrate biometrics, A-CDM and apps
- ✓ Smart contracts promise automation but adoption is early

EXIT TICKET Write 1 thing you learned today + 1 question you still have.

NEXT MISSION

Topic 5.12 — Ethical Considerations and Challenges

Thank you · Questions?